

2026.1 Multicore Computing, Project #2

(Due : 11:59pm, May 12)

Submission Rule

1. Create a directory "proj2".
2. In the directory "proj2", create sub-directories "prob1", "prob2", and "prob3".
 - 2-1. Insert into "prob1" directory: (i) JAVA source code "ManualThreads.java", (ii) a PDF or TXT document file that contains two examples of execution results (output), (iii) a demo video file (.mp4 format) that shows compilation and execution results of your source files (file size<10MB)
 - 2-2. Insert into "prob2" directory: (i) JAVA source code "ThreadPool.java", (ii) a PDF or TXT document file that contains two examples of execution results (output), (iii) a demo video file (.mp4 format) that shows compilation and execution results of your source files (file size<10MB)
 - 2-3. Insert into "prob2" directory: (i) JAVA source code "VirtualThreads.java", (ii) a PDF or TXT document file that contains two examples of execution results (output), (iii) a demo video file (.mp4 format) that shows compilation and execution results of your source files (file size<10MB)
3. zip the directory "proj2" into "proj2.zip" and submit the zip file into eClass homework board.

You are given a single-threaded Java source code (**SingleThread.java**: accessible from our class webpage) for web page analyzer. The program reads a file (**urls.txt**: accessible from our class webpage) containing URLs, download each page, compute simple statistics, and print a summary report.

execution example)

```
javac SingleThread.java
java SingleThread urls.txt
```

```
URL: https://yale.edu
```

```
Status: 200
```

```
Time: 301 ms
```

```
Characters: 98798
```

```
Lines: 1158
```

```
'java' count: 17
```

```
...
```

```
=== Summary ===
```

```
Success: 16
```

```
Failed: 0
```

```
Avg Time: 560.5625 ms
```

```
Total Characters: 3113780
```

```
Total 'java' count: 153
```

[Problem 1] Convert the given Java program into a multi-threaded version (**ManualThreads.java**) using manual thread creation. Your program should accept two command-line arguments: URL file name and the number of threads to create. You should submit (i) Java Program code, (ii) PDF or TXT file containing output example, and (iii) demo video file (.mp4 format). See the details in the submission rule.

[Problem 2] Convert the given Java program into a multi-threaded version (**ThreadPool.java**) using a fixed-size (e.g. 4) thread pool. Your program should accept two command-line arguments: URL file name and the number of threads to create. You should submit (i) Java Program code, (ii) PDF or TXT file containing output example, and (iii) demo video file (.mp4 format). See the details in the submission rule.

[Problem 3] Convert the given Java program into a multi-threaded version (**VirtualThreads.java**) using virtual threads. Your program should accept two command-line arguments: URL file name and the number of threads to create. You should submit (i) Java Program code, (ii) PDF or TXT file containing output example, and (iii) demo video file (.mp4 format). See the details in the submission rule.